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Pressure drop reduction of the impeller spiral static mixer design enabled by additive manufacturing

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ABSTRACT

A new spiral static mixer (SSM) design, also known as helical or twisted tape mixers, called the Impeller SSM, inspired by centrifugal pump impeller blades and fabricated by additive manufacturing (AM), is presented. The pressure drop and mixing efficiency of an Impeller SSM are compared to the standard SSM. Both the pressure drop and mixing characteristics are measured experimentally and validated by computational fluid dynamics (CFD) analyses. Five Impeller SSM designs of different taper, release, and attack angles were studied. Compared to a standard SSM of the same size, the Impeller SSM demonstrated a pressure drop (and power) reduction up to 18.2%. Such reductions could lead to significant energy saving and size reduction in static mixer industrial applications. Experimental results also show the ability of AM to fabricate the custom Impeller SSM without using costly machining and joining techniques.

1. Introduction

Mixing is an industrial process to create a homogenous blend for applications such as chemical reactors and heat exchange [1]. Some standard equipment for mixing include magnetic stirrers, static mixers, impellers, rotor-stator mixers, impinging jets, and manual stirring [2]. Static mixing, commonly used in chemical and food processing, utilizes pipes with stationary internal elements or structures that induce flow in directions transverse (radial and tangential) to the primary (axial) flow direction to enhance material and thermal homogenization [3]. The advantages of static mixing over other mixing methods include the lack of moving parts and associated seals, continuous process, small footprint, short mixing time, scalability, and high repeatability; these are all desirable properties for processes that require continuous production in a compact space [2].

Five of the most common types of static mixers are the: 1) spiral static mixers (SSM), also known as helical or twisted tape mixers, such as the Kenic KM™ (Chemineer, Dayton Ohio, USA) 2) corrugated static mixers such as the Koch-Sulzer SMV™ (Sulzer, Winterthur, Switzerland), 3) guide vane static mixers such as the Koch-Glitsch SMX™ (Sulzer, Winterthur, Switzerland), 4) crossed elliptical mixers such as the Komax™ mixer (Komax Systems Inc, Huntington Beach California, USA), and 5) vortex mixers such as the Chemineer HEV™ (Chemineer, Dayton Ohio, USA) [2,4–6]. The SSM is widely used in industry and an active area of research [2,4–6]. Advantages of SSM are: 1) its versatility in the types of mixing for both laminar and turbulent flows of liquid–liquid, and liquid–gas, liquid–solid [2,4,7], w) the low-pressure drop in a given length in comparison to other types of static mixers, such as the SMV™, SMX™, and Komax™ [2,4,7], and 3) the degree of mixing is a function of the length of the mixer [2,7]. Studies on mixer performance have led to an extensive data set for the pressure drop and mixing of various static mixer designs [2,4,8–10]. Shear thinning fluids also have a significant effect on the performance of static mixers [6,11–14]. Heywood et al. [8] is an early example of investigations into pressure drop and mixing of SSM, SMV, SMX, and other types of static mixers. Shah and Kale [10] correlated the pressure drop of SSM and SMX mixers with the power law model for shear thinning fluids. Belhout et al. [11] used computation fluid dynamics (CFD) to validate literature data that used the power law model for shear thinning fluids and common static mixer parameters to predict pressure dorps in SSM mixers. Rahmani et al. [12] showed that SSM mixers were equally effective mixing both Newtonian and non-Newtonian fluids. Lie et al. [13] did the same for SMX static mixers.

CFD has demonstrated its capability to accurately predict pressure drop and mixing of static mixers in single-phase and multiphase flows using techniques such as particle tracking and volume of fluid (VOF)

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solvers [3,4,15–20]. A plethora of static mixer designs used CFD for their development. Hobbs and Muzzio [21] found that a slight reduction in the standard twist of 180° in SSM results in a significant increase in performance, such as the extent of mixing per element and energy efficiency. Cheng et al. [22] applied CFD to study static mixer designs that increased cell circulation between light and dark zones in a photobioreactor. Soman [23] showed that the inclusion of small perforations in an SMX™ style mixer design leads to an improvement in the reaction rate of the homopolymerization of acrylamide. Zhang et al. [24] designed a toroidal helix pipe that achieves mixing without any internal elements.

Advancements in additive manufacturing (AM) has enabled the creation of new features in novel static mixer designs. For example, Armbruster et al. [18] utilized AM to fabricate eight new static mixer geometries with turbulence promoters to increase turbulent eddies and prevent fouling of membrane filters. Li et al. [25] applied AM to fabricate a high flow rate microfluidic mixing chamber for the extraction of copper from a sulfite solution. Nguyen et al. [26] studied AM to build an SMX™ style mixer made of nickel so that the static mixer could not only mix but act as a catalyst for chemical processes reducing the size of a tubular production reactor. These recent studies demonstrate that AM enables the creation of novel static mixers that includes features not typical in standard static mixers.

It is the goal of this study to create new features for the SSM static mixer design enabled by AM and using geometry inspired by centrifugal pump impeller geometry, referred to as the Impeller SSM. The SSM static mixer design's wide acceptance in industry and the versatility of mixing applications make it a natural starting point for improvement by AM [2]. In the following sections, the design features, CFD analysis, and experimental validation of the Impeller SSM are discussed.

2. SSM design

2.1. Standard SSM design

A standard SSM, as shown in Fig. 1(a), is characterized by stacking alternating clockwise (CW) and counterclockwise (CCW) helical elements. In Fig. 1(b), each helical element is offset by 90° and interfaces at points B_1, A_2, B_2, A_3, etc. The CW and CCW helical elements are illustrated in Fig. 1(c) and (d), respectively. Alternating CW and CCW elements of the standard SSM causes the fluid flow to repeatedly split and recombine following a baker transformation, which is a mapping operation that cuts and recombines an arbitrary space, always resulting in a chaotic distribution [27]. The number of layers of two fluids increases exponentially until the fluid is homogeneously blended by passing through the alternating CW and CCW helical elements in the SSM [2]. The repeated splitting and recombination of the fluid flow is the primary mixing mechanism for SSM [2,5,11]. As shown in Fig. 1, four parameters define the SSM are: 1) helix diameter, d, 2) helix length, l, 3) helix thickness, t, and 4) overall length, L. The standard geometry for each helical element is to have the helix length, l, to be approximately 1 to 1.5 times the diameter, d, with a pitch $p = 2l$ [2,28]. The standard thickness of a helix is not specified but should be as thin as possible. In Fig. 1(c) and (d), the CW and CCW helical elements are shown cut from a full helix, which is a helix with a pitch p. Additionally, the origin of each helix is denoted as point A, and the end of each helix along the z-axis is labeled as point B. An isometric view of the SSM can be seen in Fig. 1(b), showing the A-B interfaces of each helix in the Standard SSM.

The surfaces at points A and B along the helix centerline are flat surfaces perpendicular to the fluid flow in the SSM, as shown in Fig. 1. These flat surfaces abruptly interrupt the flow of fluid within the SSM, increasing drag within the SSM and reducing the SSM efficiency.

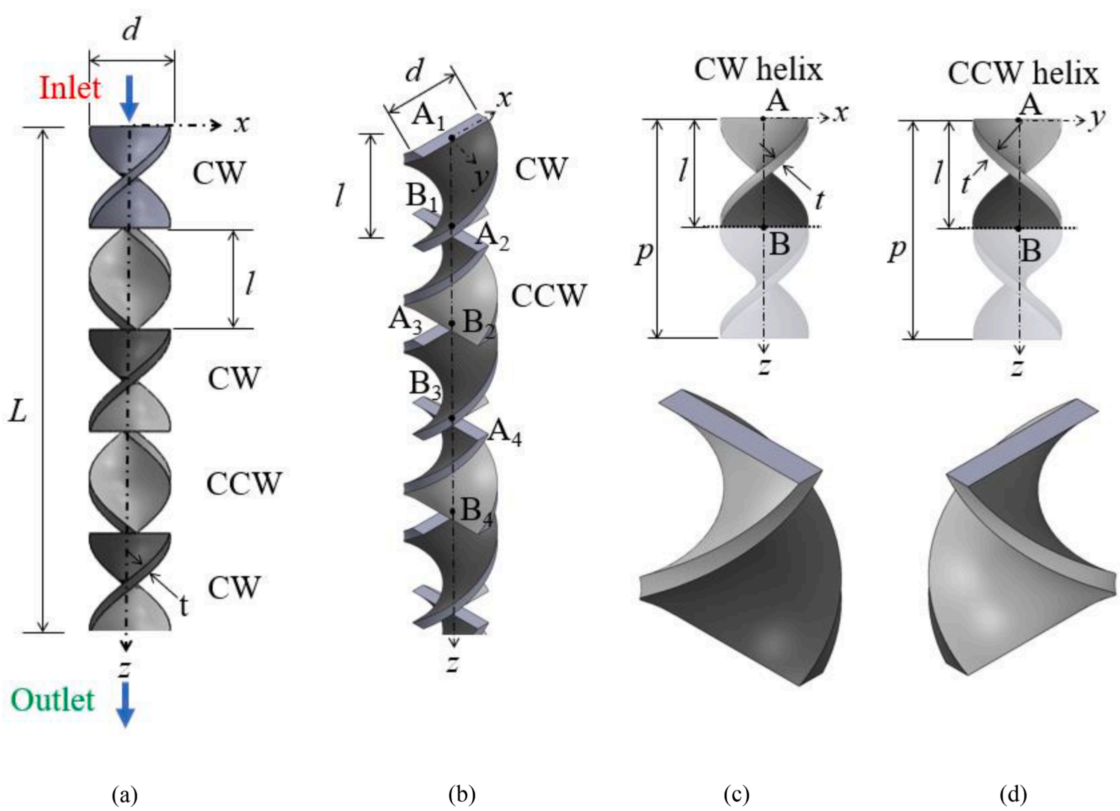

Fig. 1. A standard SSM with alternating clockwise (CW) and counter-clockwise (CCW) helix rotations with four parameters, 1) helix diameter, d, 2) helix length, l, 3) helix thickness, t, and 4) overall length, L: (a) the side and (b) perspective view of a standard SSM with alternating CW and CCW helical elements and (c) the CW and (d) CCW helix elements with point A at the top of the helix and point B at the bottom of the helix on the z-axis.

2.2. Impeller SSM design

The Impeller SSM design, as shown in Fig. 2, improves the SSM performance by reducing the abruptness of the fluid transition between each helical element. In the SSM design the fluid flow hits the flat top side of the helical element at each element and then has to quickly fill the void at the end of the helical limit. In contrast the Impeller SSM helical design is intended to cut through the flow of the fluids with its bladed top of the element and gradually transition the fluid flow through to the next element with its release angle. The core concept for the Impeller SSM came from centrifugal pumps where reduction of impeller blade angles to less than 90° with respect to the flow direction has been shown to reduce losses [29]. Centrifugal pumps have impellers with curved blades that have a similar twist shape to the helixes in an Impeller SSM. For centrifugal pumps, the fluid is rotated around a center axis by the spinning impeller. The behavior is very similar to SSM's where fluid flows around the central axis of the helix elements. The Impeller SSM introduces the three new parameters, 1) taper angle, α, 2) release angle, β, and 3) attack angle, γ, to create the taper inlet, release winglet, and attack angle features on the SSM helix element and reduce the resistance to the fluid flow.

The application of the Impeller SSM parameters happens in two stages. The first stage creates the intermediate Impeller SSM helix element, which is derived from the helix elements of standard SSM in Fig. 1. The helix created by cuts along planes P-P and Q-Q in Fig. 2 defines an intermediate CW helix element of the Impeller SSM. Fig. 2(a) shows planes P-P and Q-Q that cut a CW Impeller SSM helix element and create the taper inlet at the entrance and the release winglet at the exit, respectively. Plane P-P intersects point A and is used to define α, which is the angle between the P-P and x-y planes. In Fig. 2(b) and (c), the green surface indicates the taper inlet. Plane Q-Q intersects the z-axis at point B and is used to define the release winglet. The release angle, β, is the angle between the Q-Q and x-y planes. In Fig. 2(b) and (c), the red surface indicates the release winglet. Using this planer definition for the taper inlet and release winglet preserves the helix twist while creating an angled surface relative to the fluid flow reducing drag within the SSM.

The second stage of Impeller SSM application is the application of the attack angle, γ, to sharpen the taper inlet and release winglet of the intermediate Impeller SSM helix to reduce the flow resistance. The vertex of the taper inlet is on point A of the intermediate helix element, as shown in Fig. 3(a). For the taper inlet, in the y-z plane, triangles ACD and AGH are defined on the side of inner and outer edges ef and ij, respectively. The angle between lines AD and AC (along the y-axis) is γ in triangle ACD. Similarly, the angle between lines AH and AG (along the y-axis) is also γ in triangle AGH. Triangles ACD and AGH are swept along edges ef and ij to triangles A'C'D' and A'G'H', as shown in Fig. 3(b) and (c), respectively, to complete the creation of a sharp taper inlet shown in Fig. 3(d). The same procedure is repeated for the release winglet using point B as the vertex, Fig. 3(e). The sharpened release winglet of the CW Impeller SSM helix elements is shown in Fig. 3(f).

Both CW and CCW Impeller SSM helix elements are constructed using this method. The completed Impeller SSM consists of alternating CW and CCW elements with a 90° offset, as shown in Fig. 4. The sharp taper inlet separates the fluid flow, as shown in Fig. 4(a), reducing the resistance of the fluid transfer from helix to helix within the SSM, reducing the pressure drop of the Impeller SSM. These elements still allow for the repeated splitting and recombination of the fluid flow that is the primary mixing mechanism of the SSM static mixer design and the changes are expected to have no affect on mixing or a reduction in length of the static mixer [27]. The taper inlet and release winglet are colored as teal and purple, respectively. The Impeller SSM shown in Fig. 4(b) uses the same base dimensions, d, l, t, and L of a standard SSM and are

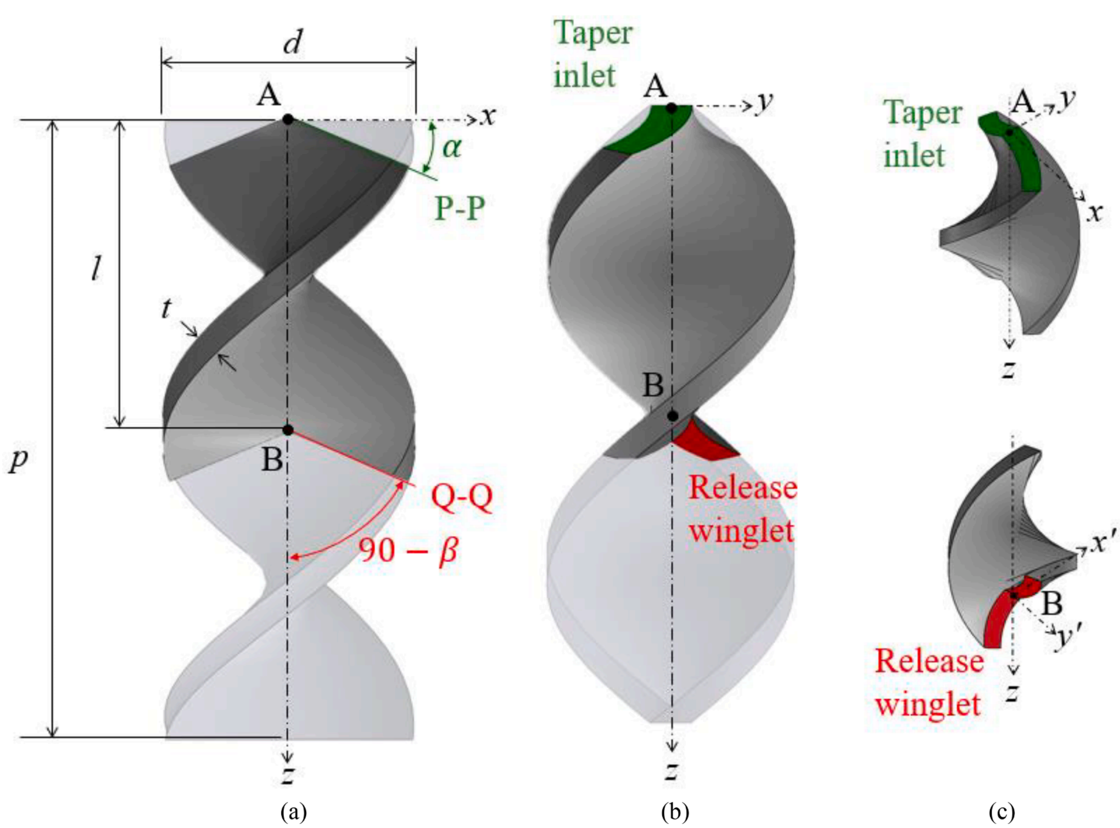

Fig. 2. (a) Front view of the Intermediate helix element created by cuts from P-P and Q-Q planes. The P-P plane intersects the z-axis at point A, has an angle $\boldsymbol{\alpha}$ with respect to the x-y plane, and is perpendicular to the x-z plane. The Q-Q plane intersects the z-axis at point B, has an angle $\boldsymbol{\beta}$ with respect to the x-y plane, and is perpendicular to the x-z plane. (b) Side view of the Intermediate helix where taper inlet is shown in green and release winglet is shown in red and (c) isometric views of the intermediate helix element showing more detail of the taper inlet and release winglet.

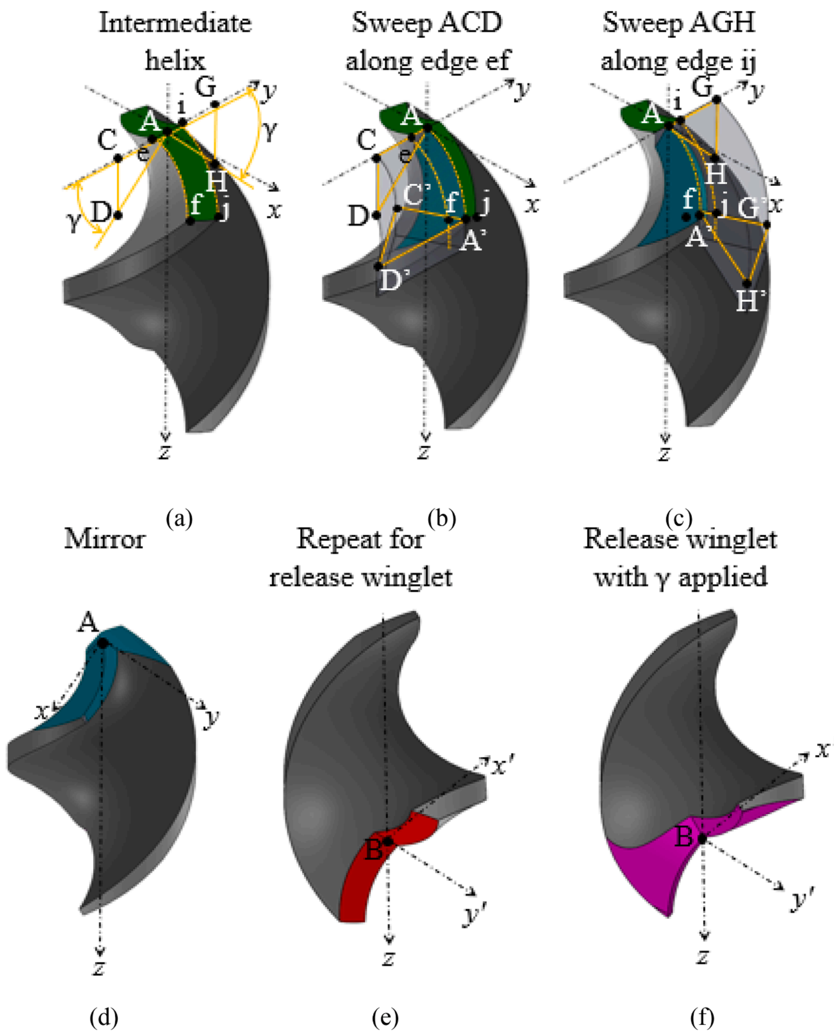

Fig. 3. (a) Isometric views of the Intermediate helix element showing the attack angle, γ, and vertex points A and B, (b) the first sweep of triangle ACD along the edge ef to triangle A'C'D' of the Intermediate helix element, (c) the second sweep of triangle AGH along the outer edge ij to triangle A'G'H' of the Intermediate helix element, (d) the sharpened taper inlet of the intermediate helix, (e) the release winglet and vertex point B used for the origin for application of the attack angle, and (f) the sharpened release winglet of the Impeller helix element.

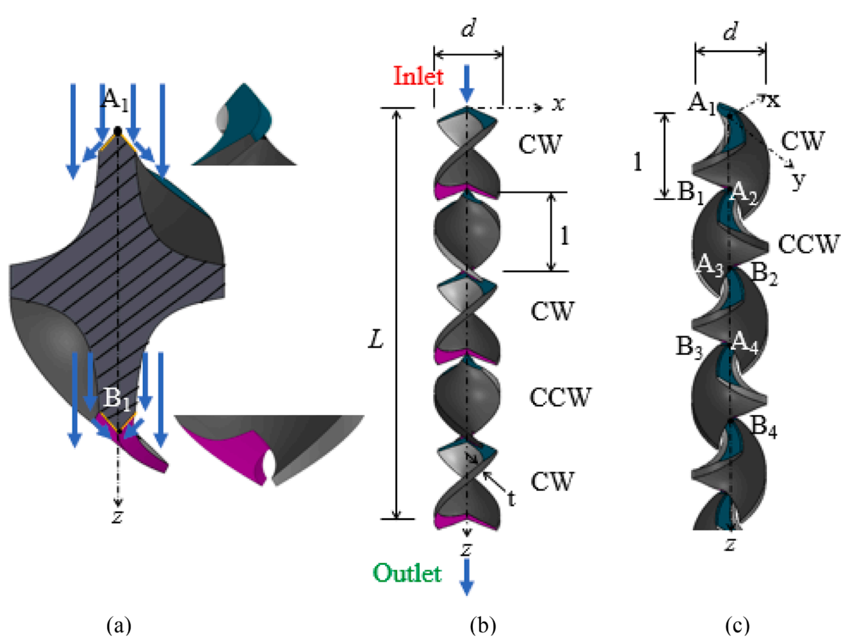

Fig. 4. (a) Cross-sectional view showing the taper inlet and release winglet gradually diverting the flow and reducing drag, the close-up views of the sharpened taper inlet and release winglet. (b) The side view of the Impeller SSM with the same d, l, t, and L as the Standard SSM shown in Fig. 1, and $\alpha = 22.5^\circ$, $\beta = 22.5^\circ$, and $\gamma = 22.5^\circ$. (c) The isometric view of the Impeller SSM.

assembled in the same manner as the standard SSM with $\alpha = 22.5°$, $\beta = 22.5°$, and $\gamma = 22.5°$. The isometric view of the Impeller SSM with five helix elements is shown in Fig. 4(c).

Each new feature of the Impeller SSM may reduce the losses associated with the fluid flow transitioning from helix to helix. These new features would be extremely difficult to fabricate using traditional machining, forming, or molding techniques, especially at the small sizes presented in this study. Most of this difficulty is from the undercuts created by the taper inlet and release winglet and the fine feature size of the attack angle. The undercuts and fine feature would require the use of complex tooling and extensive machining operations [30]. Using AM, the Impeller SSM design can be produced quickly and with the freedom to vary the design parameters as needed.

The Impeller SSM has the same *d, l, t,* and *L* as the standard SSM while α, β, and γ are varied through various configurations for testing to explore their effects on the pressure drop within the Impeller SSM. The various configurations of the Impeller SSM, referenced by a α-β-γ (in the unit of degrees) designation, are shown in Table 1 and compared in Fig. 5.

3. Static mixer pressure drop and length

Two important parameters that indicate static mixer performance are the length required for adequate mixing and pressure drop over that length. This section presents the characteristic equations to estimate the required length and associated pressure drop. Additionally the Reynolds number for the static mixer is calculated using the design parameters.

3.1. Pressure drop

The pressure drop created by the static mixer, Δp_{sm}, is calculated using the static mixer length parameter, K_L.

$$K_L = \frac{\Delta p_{sm}}{\Delta p} \tag{1}$$

where Δp is the pressure drop in an open pipe with the same d and L as that of the static mixer.

The pressure drop in an open pipe during laminar flow is expressed as [2]:

$$\Delta p = \frac{32}{Re}\frac{L}{d}\rho u^2 \tag{2}$$

where ρ is the fluid density, u is the average cross-sectional fluid speed, $Re = \rho u d/\mu$ is the Reynolds number of the flow, and μ is the fluid viscosity.

Using Eqs. (1) and (2), the pressure drop of the fluid flow for a static mixer is:

$$\Delta p_{sm} = 32 K_L \frac{\mu L u}{d^2} \tag{3}$$

For a standard SSM, K_L has experimentally been determined to be 6.9 [2].

If the fluid is shear-thinning, the effective viscosity of the fluid, μ_{eff}, for Re can be calculated using the shear rate $G^{'}$ of the static mixer and the fluid power law viscosity model [29]. For a static mixer, the shear rate can be approximated by the following [2]:

$$G^{'} = K_g \frac{u}{d} \tag{4}$$

where K_g is the static mixer shear rate coefficient, shown to be 28 in a Kenics SSM [2].

The power law viscosity model using $G^{'}$ becomes [29]

$$\mu_{eff} = kG^{'N-1} \tag{5}$$

where k is the fluid consistency index and N is the flow behavior index.

The pressure drop in a static mixer predicted by Eq. (3) does not take into consideration effects such as surface roughness or manufacturing variation but has proven to be within 15% of the actual pressure drop [2].

3.2. Mixer length

The length of a static mixer is determined by the distance needed to achieve an adequately mixed fluid. The coefficient of variation, CoV_r, is a measure of how homogeneous the fluid mixture is. For a static mixer, the CoV_r is [2]

$$CoV_r = K_i^{L/d_h} \tag{6}$$

where K_i is the blending parameter and d_h is the hydraulic diameter of the static mixer. In an SSM, K_i has experimentally been shown to be 0.87 [2]. The amount of volume that a static mixer occupies in relation to the pipe volume will change the mixing behavior of the static mixer [2]. The use of d_h accounts for this by adjusting the effective size of the static mixer for the purposes of determining its length.

There are many different measures for mixing but for static mixers CoV_r is often used because it relates directly to mixer design, as shown in Eq. (6), and amount of mixing is directly related to the position along the length of the mixer [2,5,6,11,14,31,32]. The exact CoV_r depends on the application, 0.05 is generally considered well mixed, but in more demanding mixing applications, such as mixing for color consistency, a lower CoV_r value of 0.01 should be used [2].

The definition of d_h [29] is

$$d_h = 4\frac{A_c}{P_c} \tag{7}$$

where A_c is the cross-sectional area of the static mixer and P_c is the cross-sectional wetted perimeter.

Based on Eqs. (6) and (7), the minimum static mixer length needed to achieve an effectively homogeneous fluid mixture is:

$$L = 4\frac{A_c}{P_c}\frac{\log CoV_r}{\log K_i} \tag{8}$$

3.3. Mixer Reynolds number

The mixing behavior in a static mixer is heavily influenced by the flow being either laminar or turbulent. All the correlations given by Eqs.

Table 1
SSM parameters and dimensions used in this study.

Mixer	L (mm)	d (mm)	l (mm)	t (mm)	α (degree)	β (degree)	γ (degree)
Standard SSM	53	3.0	3.75	0.5	0	0	0
15-15-22.5	53	3.0	3.75	0.5	15.0	15.0	22.5
22.5-22.5-22.5	53	3.0	3.75	0.5	22.5	22.5	22.5
30-30-22.5	53	3.0	3.75	0.5	30.0	30.0	22.5
45-45-22.5	53	3.0	3.75	0.5	45.0	45.0	22.5
67.5-67.5-22.5	53	3.0	3.75	0.5	67.5	67.5	22.5

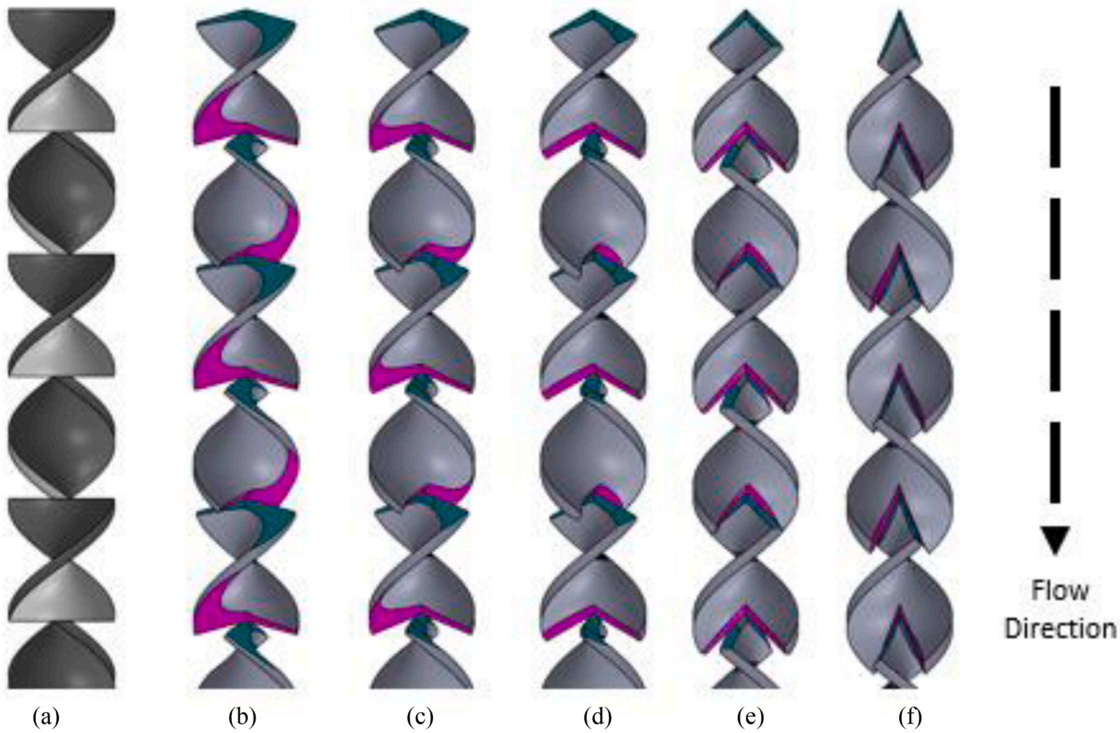

Fig. 5. The (a) standard SSM, (b) 15-15-22.5, (c) 22.5-22.5-22.5, (d) 30-30-22.5, (e) 45-45-22.5, and (f) 67.5-67.5-22.5 Impeller SSM.

(1)–(8) assume that the fluid flow is laminar as determined using μ_{eff} and d_h from Eq (5) and (7) to estimate the Re for the SSM using:

$$Re = \frac{\rho u d_h}{\mu_{eff}} \tag{9}$$

4. Design and additive manufacturing of standard and impeller SSMs

The standard SSM design, Fig. 6(a), used for this study has a $d = 3$ mm, $l = 3.75$ mm, and $t = 0.5$ mm, all of which are within the design guidelines for a standard SSM [28]. The d and t values result in $A_c = 5.58$ mm^2 and $P_c = 14.34$ mm as determined by the 3D model of the SSM element. The A_c and P_c with a target $CoV = 0.01$ and $K_i = 0.87$ are used in Eq. (8) to determine $L = 53$ mm (14 helical elements) for a mixing application that needs color consistency. In Fig. 6(b), the 22.5–22.5–22.5 Impeller SSM is shown with $\alpha = 22.5°$, $\beta = 22.5°$, and $\gamma = 22.5°$ (as described in Fig. 4). To allow for experimental measurement of pressure drop, the SSM also has two M5 threaded pressure sensor ports placed at 7 mm and 37 mm from the start of the SSM, as shown in Fig. 6(c). Fig. 6(d) shows the internal cut-out of a standard SSM manufactured on a stereolithography (SLA) AM machine (Form 2, FormLabs Somerville Massachusetts, USA) with two pressure sensors installed. All SSMs in this study were printed using SLA resin (Clear v4, FormLabs Somerville Massachusetts, USA), a layer height of 0.05 mm, and post processed using the Form Wash and Cure (FormLabs Somerville Massachusetts, USA) post processing equipment following the manufacturers recommended procedures for the SLA resin. The internal cut-out of a 22.5-22.5-22.5 Impeller SSM can be seen in Fig. 6(e) showing the fine features produced by AM. The size and dimensions of the AM parts were validated using a caliper (8000-F6, Products Engineering Corporation, Torrance California, USA).

Both the standard SSM and Impeller SSM configurations shown in Fig. 6 require approximately 7 mL of resin to be printed on the SLA AM

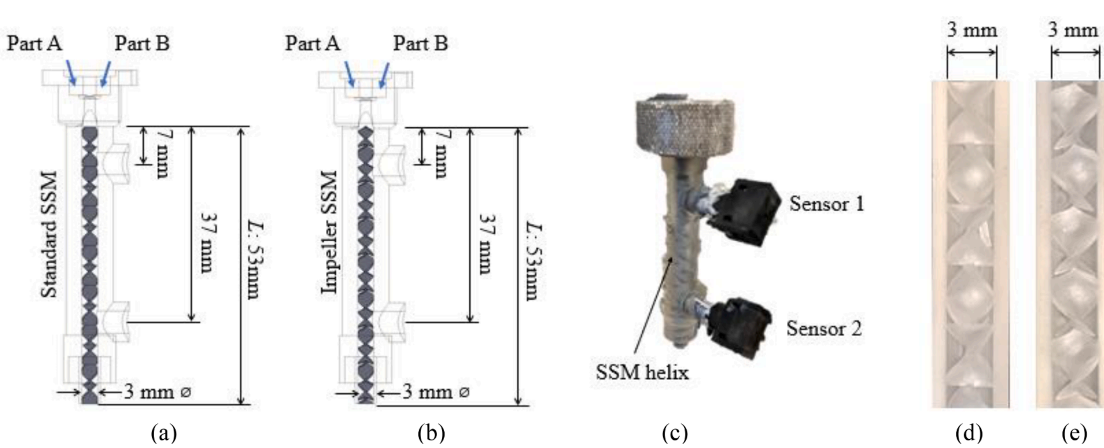

Fig. 6. (a) The standard SSM and (b) 22.5-22.5-22.5 Impeller SSM with the location of the pressure ports and overall length annotated, (c) the standard SSM produced using SLA and with pressure sensors installed, and cutaway view of the (d) standard SSM and (e) 22.5-22.5-22.5 Impeller SSM produced using SLA.

machine. At the time of writing 1l of resin costs $149, making the total material cost per SSM is approximately $1.05. This makes the cost of producing a SSM comparable to purchasing commercially available static mixers of comparable size and length.

5. CFD modeling of standard SSM and impeller SSM

CFD modeling was used to accurately simulate the pressure drop of the standard SSM and Impeller SSM. Using CFD will provide a detailed analysis of the fluid flow within the static mixers that cannot be easily obtained experimentally. The Fluent™ (v19.1) CFD software package from ANSYS (Canonsburg Pennsylvania, USA) was used to simulate the fluid flow in the standard SSM and various Impeller SSM configurations.

5.1. CFD mesh, boundary conditions, and initial conditions

The mesh for the CFD was generated from the 3D CAD model of the standard SSM and Impeller SSM configurations, by using a Boolean subtract on the 3D model to isolate the fluid regime within the 3D model. In Fig. 7(a), the 3D CAD models for a standard SSM and a 22.5-22.5-22.5 Impeller SSM are shown. In Fig. 7(b), the fluid domain inside both SSMs is generated from a Boolean subtract operation. Fig. 7(c) shows the tetrahedral mesh generated on the fluid domain by the ANSYS meshing tool using the patch conforming method [33] with a mesh size of 0.2 mm, resulting in 6.6 × 10ˆ6 meshing elements and was checked for independence. Due to the size of the model relative to the mesh size, only a small portion of the mesh is shown for visualization purposes.

The mesh independence stud for the standard SSM can be seen below in Table 2 where among the tested configurations, the coarsest mesh size was 0.8 mm which failed to converge on a solution. The steps in element size were chosen such that the total number of meshing elements was at least double the previous mesh. As the meshing element size was reduced the change in solution value was recorded. The 0.2 mm element size was chosen because the percentage change in the answer was within 1% of the 0.1 mm element size solution and any further refinement of the CFD mesh was providing diminish returns at the expense of extra computation time.

The boundary conditions of the CFD model are assumed to be a constant volumetric flow inlet, gauge pressure outlet, and non-slip walls. The fluids used in the CFD models are made up of two parts, Part A and Part B. Each fluid part has a separate constant volumetric flow inlet, as shown in Fig. 6(a) and (b). The Volume of Fluid (VOF) model [17] was used for the multiphase flow of both the Part A and Part B components flowing through the static mixer. The material properties used for the CFD model are discussed in the next section. The VOF model was run using the explicit scheme, a courant number of 0.25, and two Eulerian phases. The solver was pressure based using the “Fractional Step” scheme for the pressure–velocity coupling, “Least Squares Cell” for the spatial discretization gradient, “PRESTO!” for the pressure spatial discretization gradient, “QUICK” for the momentum spatial discretization gradient, “Geo-Reconstruct” for the spatial discretization gradient, and the “Non-Iterative Time Advancement” and “First order Implicit” for the Transient Formulation. A transient time step of 1×10^{-6} s was used and the pressure drop at ten seconds (10^7 time steps) was considered steady state. The time step was chosen empirically so that the continuity residual for each time step was below 10^{-5} and the velocity residuals where below 10^{-4} for each simulation, allowing for consistency of CFD parameters across all configurations of SSM. The determination of ten seconds as a steady state solution was also empirically chosen because the transient solution stopped changing for at least 10^4 time steps for each flow rate, a steady state fluent solution as not used because the model would not converge.. The initial conditions for the model were determined using the hybrid initialization feature of Fluent™ for a fluid [34].

The splitting and recombining of flow by the SSM elements does not cause any dispersion of the fluids in laminar flow [2,17]. Instead, the

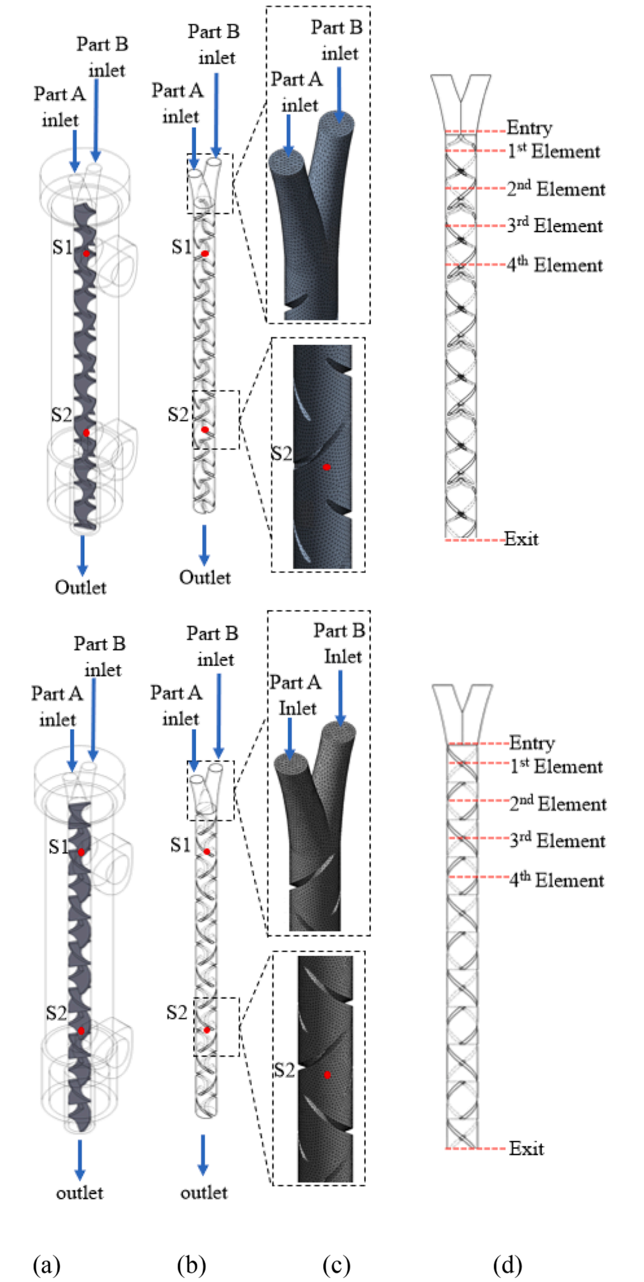

Fig. 7. (a) The 3D model and (b) the fluid domain of the CFD models for the standard SSM and 22.5-22.5-22.5 Impeller SSM showing the fluid inlets, outlet, and pressure measurement locations marked, (c) the CFD mesh for standard SSM and 22.5-22.5-22.5 Impeller SSM, and (d) the position of the cross-sections where mixing of the CFD model to be measured at the entry and middle of the first four elements for every SSM configuration.

Table 2
Mesh independence study for SSM flowrates.

Element size (mm)	Pressure drop [kPa] 1 mL/min	2 mL/min	3 mL/min	Percentage change 1 mL/min	2 mL/min	3 mL/min	Meshing elements
0.8	N/A	N/A	N/A				1.0 × 10ˆ5
0.6	352	583	771				2.4 × 10ˆ5
0.4	369	624	835	5.0	7.0	8.0	8.3 × 10ˆ5
0.2	379	642	862	2.5	3.0	3.0	6.6 × 10ˆ6
0.1	381	646	866	0.7	0.6	0.5	5.3 × 10ˆ7

layers of fluid become progressively smaller and numerous until molecular diffusion causes the fluids to become effective mixed. The VOF model was chosen to model this process due to its strength in tracking the interface of two fluids. As the fluid progresses towards the end of the SSM the layers will become small enough that they will be contained within a single meshing element. At this point the fluid is considered mixed.

The predicted pressures at points S1 and S2, which are 7 mm and 37 mm, respectively, from the start of the SSM, as indicated in Fig. 7(a), (b), and (c), are captured as output and will be compared against experimental data. In order to evaluate the level of mixing, contour plots showing the volume fraction of Part A and Part B are captured at the entry and in the middle of the first four elements as well as the exit of both the standard SSM and Impeller SSM, as shown in Fig. 7(d). The CoV_r is measured using the mean, $\mu_{volume\ fraction}$, and standard deviation, $\sigma_{volume\ fraction}$, of the volume fraction for meshing elements in the contour plot cross section.The CFD model for the standard SSM and Impeller SSM configurations were based on an alkoxysilicone sealant silicone from Dow, Inc. (Midland, MI, USA) with Part A and Part B components as the model fluids. The alkoxysilicone mixture was simulated at three volumetric flow rates: 1, 2, and 3 mL/min. The Part A and Part B components individually had a flow rate half of the overall volumetric flow rate; i.e., 0.5, 1, and 1.5 mL/min. A total of 18 CFD simulations were conducted on the three flow rates and six SSM configurations.

5.2. Fluid properties in CFD

The densities, ρ, and viscosities, μ, of the alkoxysilicone Part A and Part B components are required as inputs for the CFD model. The densities for Part A and Part B are 1350 and 1240 kg/m^3, respectively. A power law viscosity model, Eq. (5), is used to capture the shear thinning rheological characteristics of the alkoxysilicone components as shown in Fig. 8. The fitted power law equation parameters for Part A and Part B are = 190 and 1110 Pa-s^N for k and 0.7 and 0.5 for N, respectively. The alkoxysilicone components are miscible with each other. For the purposes this study the effects of surface tension was ignored because the Capillary number is about 10. All fluid properties were provided by the alkoxysilicone manufacturer, Dow Silicones Corporation.

From inspection of Fig. 8 it can be seen that Part A does not have a linear line on the log-log graph as expected for a power law fluid. The expected shear rates of this study based on Eq. (4) range from 22 to 66 s^{-1}. When the shear data is extrapolated out for the shear rates using a logarithmic fit curve the power law viscosity model overestimates the μ_{eff} of Part A by approximately 7% to 22% at the shear rates used in this study. However due to μ_{eff} of Part A being approximately a quarter of the μ_{eff} for Part B, the combined μ_{eff} used in Eq. (3) is overestimated by 2–5%.

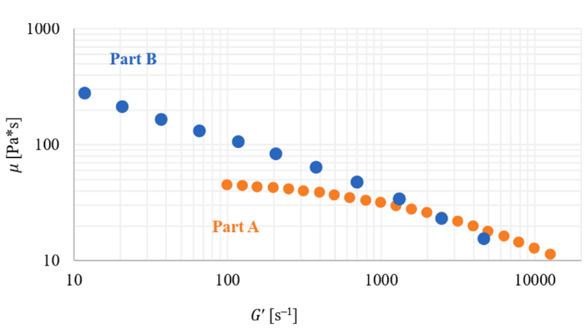

Fig. 8. Rheological data for alkoxysilicone provided by Dow, Inc.

6. Experimental setup of pressure measurement of standard and impeller SSMs

An experimental pressure drop measurement was performed to validate the CFD models of the standard SSM and various Impeller SSM configurations. The fluid flow in the standard SSM and various Impeller SSM configurations was created using a dual progressive cavity pump (Model PTC 2-SD01, Shanghai Baoling Intelligent Technology Co., Shanghai, China) shown in Fig. 9. Progressive cavity pumps are a type of positive displacement pump where a rotor forces fluid through a series of fixed cavities, which creates a precise constant volumetric flow rate for both high and low viscosity fluids [35]. These dual progressive cavity pumps, which are placed next to one another, deliver two fluids, Part A and Part B, to the SSM to replicate conditions from the CFD model.

The progressive cavity pump flow rate was calibrated by dispensing the alkoxysilicone through each static mixer for 60 s and weighing the material extruded through the mixers using an electronic scale (Model Tree HRB103, LW Measurements LLC, Rohnert Park, CA, USA). The calibrated flow rates used in all experiments are 0.99±.03, 2.02±.03, and 2.98±.03 mL/min for the target volumetric flow rates of 1, 2, and 3 mL/min respectively.

Pressure sensors (Model 24PCGFH6G, Honeywell Charlotte North Carolina, USA) are installed into the two pressure ports on the standard SSM and Impeller SSM configurations. The pressure sensors installed at the 7 mm and 37 mm locations are marked as Sensor 1 and Sensor 2, respectively, as shown in Fig. 9. The signal from the pressure sensor is

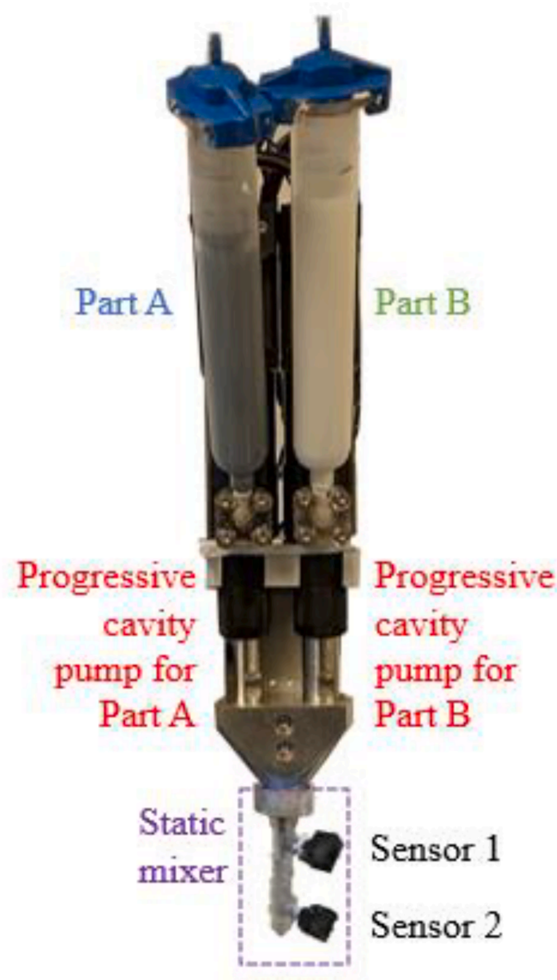

Fig. 9. The dual progressive cavity pump used for dispensing Part A and Part B at a constant volumetric flow rate. The displayed standard SSM has two pressure sensors for monitoring the pressure drop.

amplified by a custom op-amp circuit with a gain of 10 and was read by the analog input of an Arduino microprocessor (Arduino Mega 2560 Rev3) at a 30 Hz sampling rate. Both pressure sensors were calibrated using a pressure gauge (Model DPGA-07, Dwyer Instruments Michigan City Indiana, USA) and a custom pressure manifold. The pressure manifold was pressurized and the readings from the pressure sensors and gauge were taken. The calibration showed a linear response to changes in pressure and approximately 1% hysteresis in the pressure measurement, which conforms to the sensor datasheet [36]. Pressure sensors are replaced for each static mixer tested, since the fluids used in this study will begin to harden after approximately a hour after mixing, and the procedure described above is repeated for each pressure sensor used.

For all tested configurations, the progressive cavity pump was run at the calibrated target flow rate of 1, 2, and 3 mL/min for 20 s while the pressures at Sensors 1 and 2 were recorded. The steady-state pressure drop between Sensors 1 and 2 is calculated by averaging the pressure difference over the middle 15 s of the steady-state flow to eliminate transient effects caused by the starting or stopping of the flow. This experimental procedure was repeated three times for each of the setups represented by the 18 CFD simulations, for a total of 54 tests. All tests were performed at room temperature, approximately 21 C. A summary of all operating conditions of used for all experimental tests is shown below in Table 3.

7. Results

Results of the pressure drop from the experiments and CFD simulations for all SSM configurations are compared to determine the validly of the CFD results. The empirical model for the pressure drop with a standard SSM is used to validate the CFD results based on standard SSM geometry. Further analysis of the fluid velocity within each SSM from the CFD simulations will provide further insight into how the Impeller SSM parameters affect the internal fluid flow to reduce its pressure drop.

7.1. Pressure measurements and validation of CFD model

Fig. 10(a) shows the pressure data recorded from the 22.5-22.5-22.5 Impeller SSM with a 1 mL/min volumetric flow rate of alkoxysilicone at Sensors 1 and 2, as well as the pressure difference between these two sensors. This pump is switched on at the 4 s mark, and switched off at 24 s. The steady-state region used for the average pressure drop calculation is between 6.5 and 21.5 s. The pressure difference between Sensors 1 and 2 in the steady-state area is averaged to determine the pressure drop of the SSM for this test. A close up of the steady state region and the data from Sensor 2 is shown in Fig. 10(b). By only using data from the steady-state region, effects of the transients created by the progressive cavity pumps starting and stopping the flow and the residual pressure that remained within the SSM in between tests do not significantly affect the pressure measurement data.

The 22.5-22.5-22.5 Impeller SSM CFD steady-state pressure contour for the 1 mL/min flow rate simulation is shown in Fig. 11. The pressure drop prediction from the CFD simulations is the difference in pressure between points S1 and S2 indicated in Fig. 11. For the 1 mL/min flow rate, the CFD simulation yielded pressure drops of 378 and 301 kPa for the standard SSM and 22.5-22.5-22.5 Impeller SSM, respectively. This denotes a 20.3% reduction in pressure drop for the 22.5-22.5-22.5 Impeller SSM. At the same flow rate, the average experimental pressure drops were 392 and 320 kPa for the standard SSM and 22.5-22.5-22.5 Impeller SSM, respectively, giving a 18.2% reduction in pressure drop.

Table 3
Operating conditions for experimental tests.

Operating conditions	Value
Total number of tests	54
Total number ofstatic mixers tested	18
Calibrated flow rates	0.99±.03, 2.02±.03, and 2.98±.03 mL/min
Temperature	Room, approximately 21 C
Ambient pressure	Atmosphere
Pressure sensors	Model 24PCGFH6G
Microcontroller	Arduino Mega 2560 Rev3
Data sample rate	30 Hz
Signal amplification circuit	Custom Op-amp circuit with a gain of 10

The pressure drops between Sensors 1 and 2 from the experiments and between S1 and S2 in the CFD models are presented in Fig. 12 for all 6 tested SSM configurations. The error bars on the experimental values represent the range of pressure drop recorded during the experimental tests and the dotted line is the expected pressure drop in the static mixer over a 30 mm distance predicted by Eq. (3) using the d and L of the standard SSM from Table 1, the fluid properties presented in Section 5.2, and the standard SSM K_L and K_G values. The results show good agreement between the CFD and experiments on the pressure drop for all SSMs, though it is less accurate for the 1 mL/min flow rate. This is most likely due to the power law viscosity model being less accurate at this flow rate. For all standard SSM experiments and CFD models, the expected pressure drop predicted by Eq. (3) was within 15% of the actual pressure drop indicating the results conform with the expected performance of a standard SSM [2].

The error between the CFD values, and the experimental value is primally attributed to the discrepancy of the power law model for Part B. As can be seen in Fig. 8, the shear data for Part B does not fully follow the power law model. At lower flow rates which cause lower shear rates the difference between the Power law model and actual viscosity are largest, this correlates with the experimental data and CFD data having the largest differences at the 1 mL/min tests. Additionally, the spread in experimental data is attributed to standard experimental error such as slight differences in tested mixers, the 0.3 mL/min spread for the calibrated flow rates, and the expected error 1% error indicated in the pressure sensor data sheet [36].

The average, maximum, and percent pressure difference between the CFD data and experimental data can be seen below in Table 4.

The reduction in experimental measured pressure drop for each Impeller SSM configuration ranged between 1.8% and 18.2%. Overall, the 15-15-22.5 Impeller SSM configuration showed the most improvement with an average pressure drop reduction of 16.4% for three flow rates, resulting in a K_L of 5.8. The 22.5-22.5-22.5 configuration is the next best at 14.1% reduction in pressure drop resulting in a K_L of 5.9. The 45-45-22.5 and 67.5-67.5-22.5 configurations only showed marginal improvement over the standard SSM with an average of 3.3 and 2.3% reduction in pressure drop. This made the K_L for these configurations 6.7, which is not much better than the standard SSM K_L of 6.9. Reducing the pressure drop in the Impeller SSM will reduce the amount of power needed to create fluid flow.

7.2. CFD analysis of mixing and internal flow

The 15-15-22.5, 22.5-22.5-22.5 and 30-30-22.5 Impeller SSM configurations had the best experimental reduction in pressure drop compared to the standard SSM. The CFD data is compared between the standard SSM and these three Impeller SSM configurations to gain insight of mixing and flow characteristics.

Fig. 13 shows a comparison of the contour plots of the volume percent of Part A and Part B over cross sections at the mixer inlet and the center of the first four SSM elements for the standard SSM and the 15-15-22.5, 22.5-22.5-22.5 and 30-30-22.5 Impeller SSM configurations at a volumetric flow rate of 2 mL/min. The red regions are alkoxysilicone Part A, the blue is alkoxysilicone Part B, the green is fully mixed alkoxysilicone, and the intermediate colors are the other varying degrees of mixing. The plots and CoV_r calculated using Eq. (6) are shown in Table 5 below indicate a similar level of mixing performance for all four static mixers. The is a deviation in the data from the expected CoV_r at the third and fourth element which goes away when compared at the exit.

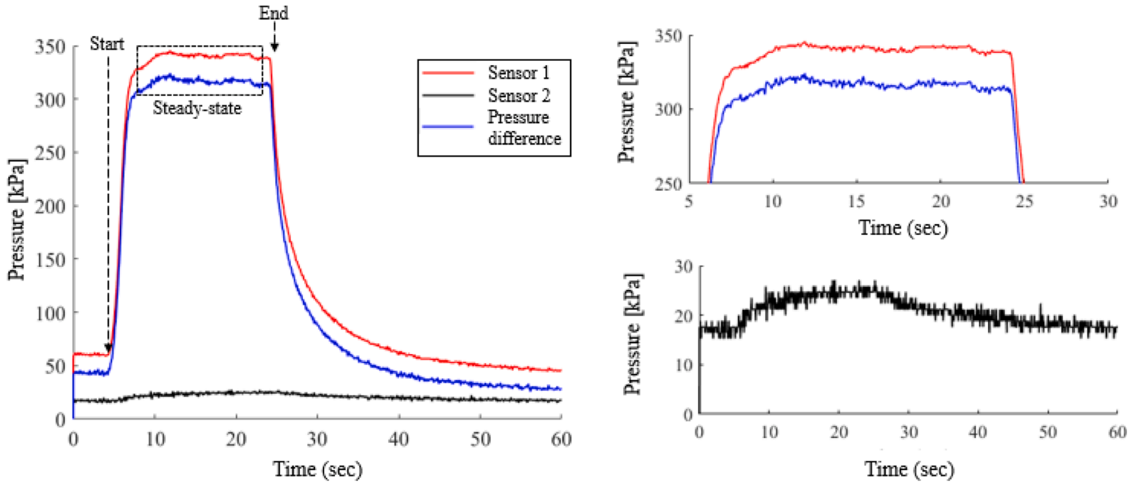

Fig. 10. (a) Pressure recorded using the Sensors 1 and 2 for 22.5-22.5-22.5 Impeller SSM with a 1 mL/min volumetric flow as well as the pressure difference between them. The test began at 4 s, ended at 24 s, and the steady-state region was between 6.5 and 21.5 s. (b) Close up of the data for steady-state region and Sensor 2.

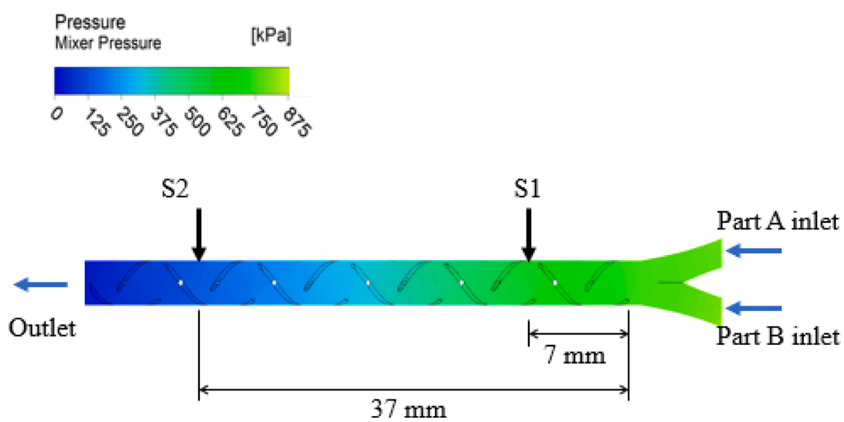

Fig. 11. CFD pressure contours for the 22.5-22.5-22.5 Impeller SSM with a 1 mL/min steady-state volumetric flow rate. The pressure drop is the difference between the pressures at S1 and S2.

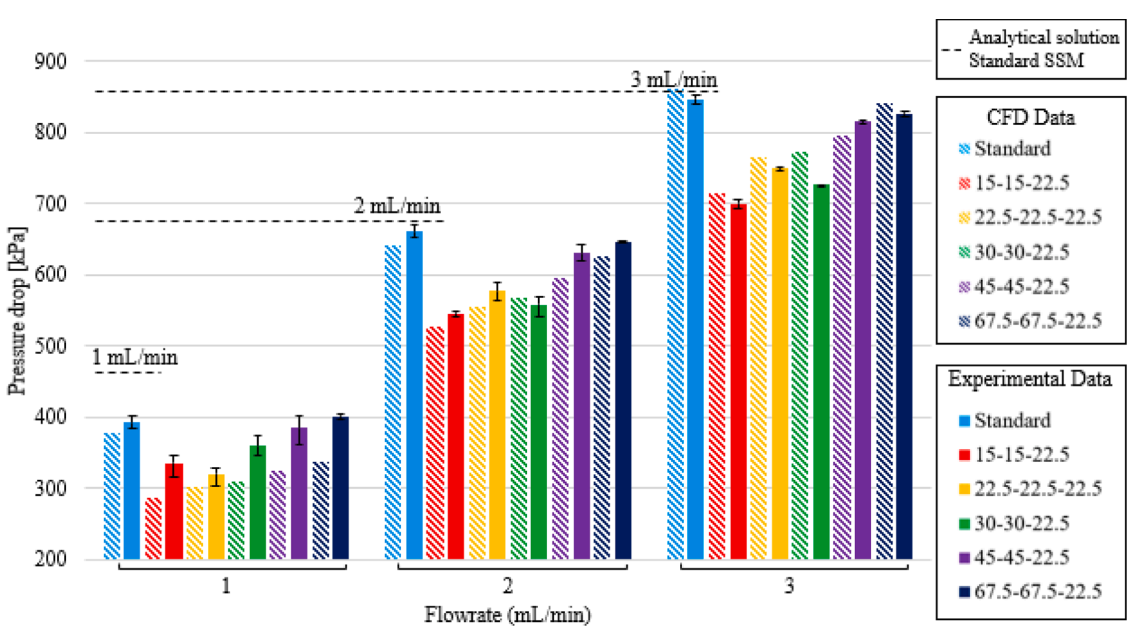

Fig. 12. The experimental and CFD pressure drops for the Impeller SSM and standard SSM at a flow rate of 1, 2, and 3 mL/min. The dashed black line is the expected pressure drop for the standard SSM predicted by Eq. (3).

Table 4
Average, maximum, and average pressure difference for each SSM configuration.

	Average difference [kPa]	Maximum difference [kPa]	Percent difference [%]
Standard SSM	15.8	18.7	2.7
150-150-225	27.6	49.3	6.7
225-225-225	19.2	22.2	4.0
300-300-225	36.0	51.1	7.5
450-450-225	37.6	59.3	7.7
675-675-225	32.9	63.0	6.9

The exact cause is unknown but could be due to how the interface between each fluid is handled in the VOF model. At the interface between two fluids the ratio between the two fluids is calculated and applied to an entire meshing element increasing the homogeneity of meshing elements, potentially causing the $\sigma_{volume\ fraction}$ to drop faster than expected.

Due to the mixing mechanism of the SSM being the splitting and recombining of fluid layers it was not expected for the ISSM to change the $K_I$value of 0.87. From Eq. (6) it is expected that the CoV_r for the SSM at the exit would be 0.01 and all CFD data was consitent with this. The same can be said for the $K_g$values of the ISSM. In laminar flow the main source of shear is the non-slip drag from the walls and mixing elements in the static mixer. The ISSM design does change the internal surface area slightly but not in any way that could be considered significant. Due to this the K_g is of the ISSM is unchanged at 28.

Fig. 14 shows the 3D velocity magnitude fields from CFD simulations run at 2 mL/min volumetric flow of alkoxysilicone for the standard SSM (Fig. 15(a)) and the three best performing Impeller SSM configurations: 15-15-22.5 in Fig. 14(b), 22.5-22.5-22.5 in Fig. 14(c), and 30-30-22.5 in Fig. 14(d). The junctions between SSM helical elements are circled to highlight the improvements that the Impeller SSM makes to the internal fluid flow patterns, compared to the standard SSM, by reducing the interferences at the helix junctions. The standard SSM flow has noticeably larger areas that have very slow to stagnant fluid flow (the blue areas) in comparison to the Impeller SSM configurations. Additionally, for all the fluid velocities in the Impeller SSM configurations, there are relatively uninterrupted streams of high velocity (red colored) flow indicating that the Part A and Part B alkoxysilicone fluids are traveling and mixing through the SSM efficiently.

A close-up isometric view of the B_2 junction of helical element shows the region at the top of the next element. For the standard SSM, Fig. 16 (a), the area has a noticeable slow fluid velocity (blue color) indicating that a significant impedance to the fluid flow exists in this area. The same region on the three best performing Impeller SSMs, Fig. 16(b)–(d), indicates relatively little fluid interference, as shown by the green color that is prominent in the B_2 junction region.

The effect of the Impeller SSM geometry on the fluid velocity can also be seen in the fluid velocity contour plots that lies on the central *x-z* plane as shown in Fig. 16. The high fluid velocity areas (green and yellow) for the15-15-22.5 Impeller SSM in Fig. 16(b), are much larger and more continuous than those of the standard SSM in Fig. 16(a). Additionally, the abrupt stop of the high velocity areas that is seen at the junction between two helical elements for the standard SSM does not occur for the Impeller SSM. This is again an indicator of the reduced impedance to the flow by the Impeller SSM helix. This behavior is also seen in the 22.5-22.5-22.5 and 30-30-22.5 Impeller SSMs in Fig. 16(c) and (d), respectively.

Inspection of the CFD data for the 67.5-67.5-22.5 Impeller SSM configuration reveals why it did not have as large of pressure drop reduction as other Impeller SSMs. The circled regions show the fluid having low velocity, indicating a stagnation point. This stagnation point is created by the extra height and twist of the release winglet getting too close to the next mixing element's taper inlet. This behavior was also noticed in the 45-45-22.5 Impeller SSM as well (Fig. 17).

Table 5
CoV_r data for the first four SSM elements.

Mixer	Entry	1st Element	2nd Element	3rd Element	4th Element	Exit
Standard SSM	0.90	0.78	0.58	0.36	0.20	0.01
Standard SSM Expected	0.87	0.76	0.58	0.44	0.34	0.01
15-15-22.5	0.90	0.71	0.46	0.34	0.19	0.01
22.5-22.5-22.5	0.84	0.79	0.42	0.35	0.20	0.01
30-30-22.5	0.87	0.72	0.65	0.35	0.20	0.01

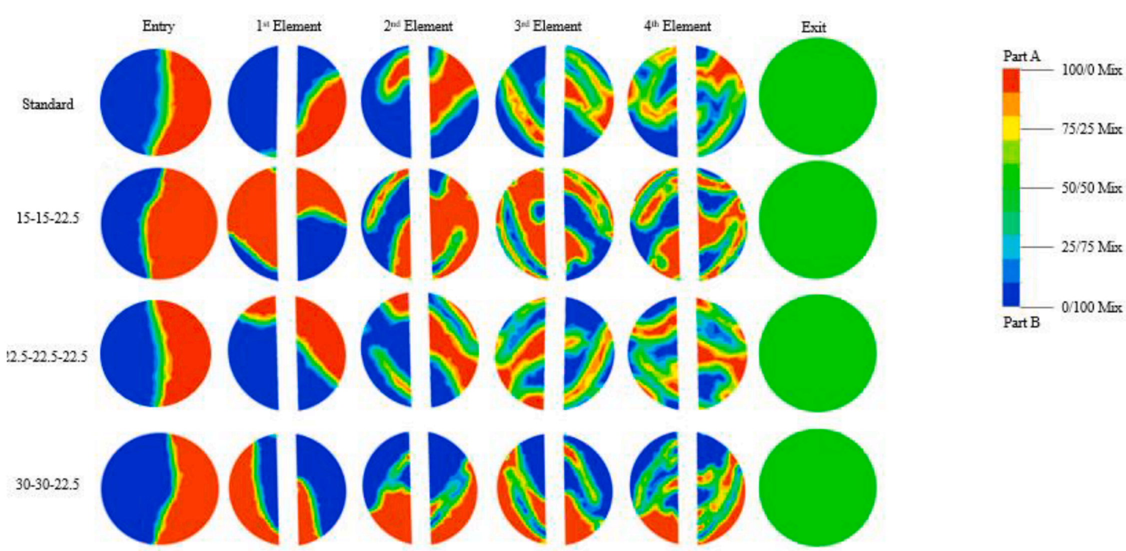

Fig. 13. Volume percent of Part A (red) and Part B (blue) components at cross sections located at the inlet, the center of the first four elements, and exit (see Fig. 7 (d)) of the standard SSM, top row, and three best Impeller SSM configurations. The green color indicates fully mixed alkoxysilicone.

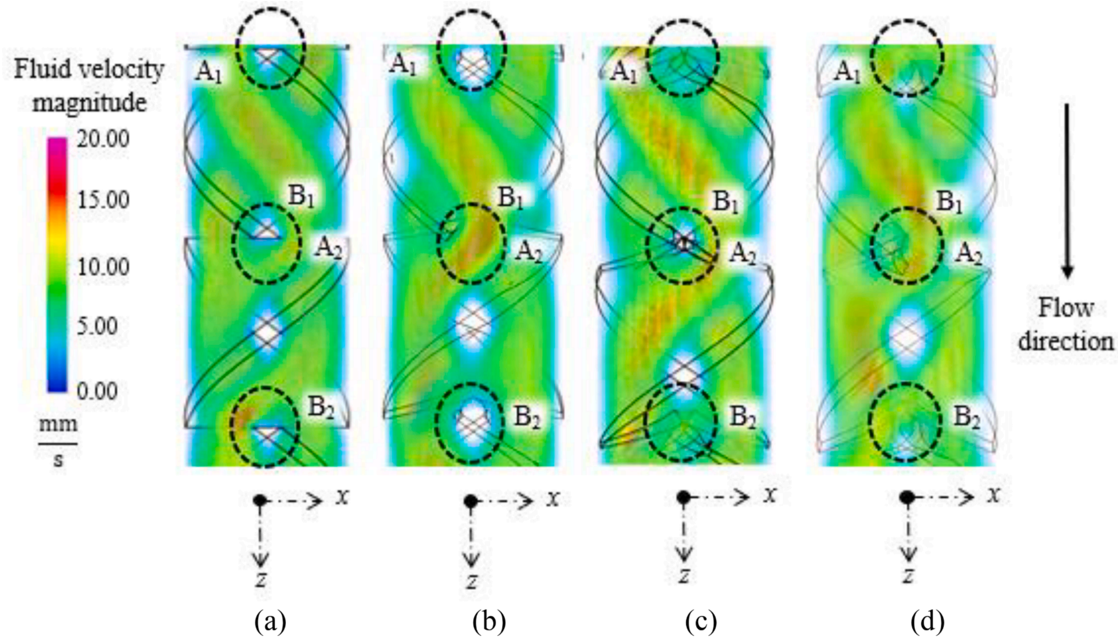

Fig. 14. Side view of the 3D velocity magnitude fields for (a) the standard SSM, and (b) 15-15-22.5, (c) 22.5-22.5-22.5, and (d) 30-30-22.5 Impeller SSMs. The circled areas indicate the junctions between two helical elements.

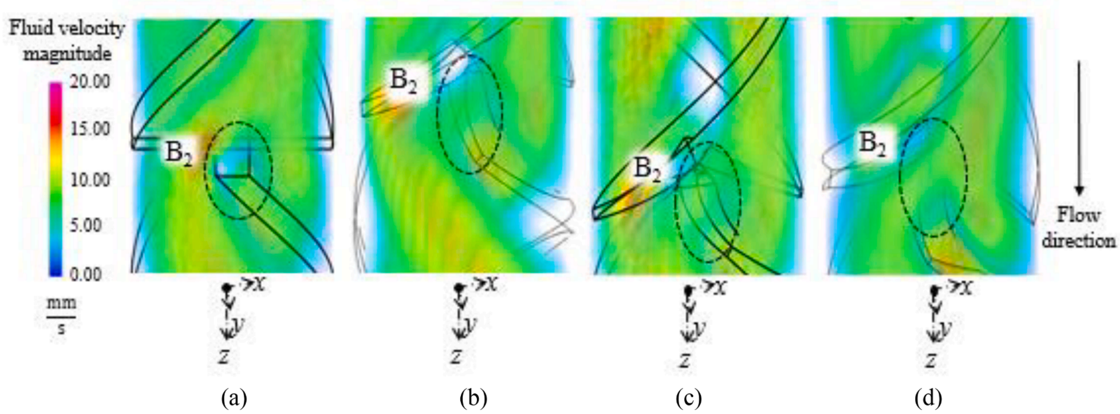

Fig. 15. Isometric view of the B_2 interface on the (a) standard SSM, (b) 15-15-22.5, (c) 22.5-22.5-22.5, and (d) 30-30-22.5 Impeller SSMs.

In general the correlations given in Eqs. (1)–(8) assume that the laminar flow exists. In the CFD data for all configurations the Reynolds number tended to be on the order of 10^{-5} for all SSM configuration and flow rates indicating that all flow tested in this study was laminar. The maximum Reynolds Number for the standard SSM with a 1, 2, and 3 mL/min volumetric flow rate are 3.15×10^{-5}, 8.32×10^{-5}, 10.38×10^{-5} respectively. This agrees with the expected Reynolds number of a standard SSM calculated using Eq. (9) where the expected Reynolds number are 3.1×10^{-5}, 8.37×10^{-5}, and 10.4×10^{-5} for the 1 mL/min, 2 mL/min, and 3 mL/min flow rates respectively.

8. Conclusions and future work

The Impeller SSM, inspired by a centrifugal pump impeller, adds advanced features to the standard SSM design, which result in a reduction in pressure drop as demonstrated both experimentally and computationally. Experiments showed that the Impeller SSM design could lower the pressure drop by up to 18.2% compared to a standard SSM design. CFD data showed improvements as high as 20%. The 15-15-22.5 Impeller SSM configuration was the experimentally tested configuration with the best performance, with a pressure drop reduction of 16.4%, averaged over three flow rates. The second best SSM configuration, 22.5-22.5-22.5, had an average pressure drop reduction of 14.1%. The 45-45-22.5 and 67.5-67.5-22.5 SSM configurations showed the least improvement over the standard SSM, with a 3.3% and 2.3% pressure drop reduction, respectively. From these pressure drop reductions the ISSM design can be shown to have a K_L of approximately 5.8 to 5.9 for its best performing configurations compared to the standard SSM K_L of 6.9 while keeping the same mixing performance. The K_L of the best performing ISSM is compared against the expected K_L of SSM and other types of static mixers in Table 6 [2].

This reduction in performance is likely due to a stagnation in flow pattern. CFD analysis show similar mixing patterns between Part A and Part B components of the Impeller SSMs compared to the standard SSM.

Due to the capability of SLA technology in producing parts with geometric variations, future work can focus on a more detailed study the effects of the design parameters of Impeller SSM on pressure drop, leading to an optimized design. This method can also be applied to the development of other novel static mixer designs. Additional efforts can also be applied to validating the mixing properties of the static mixer

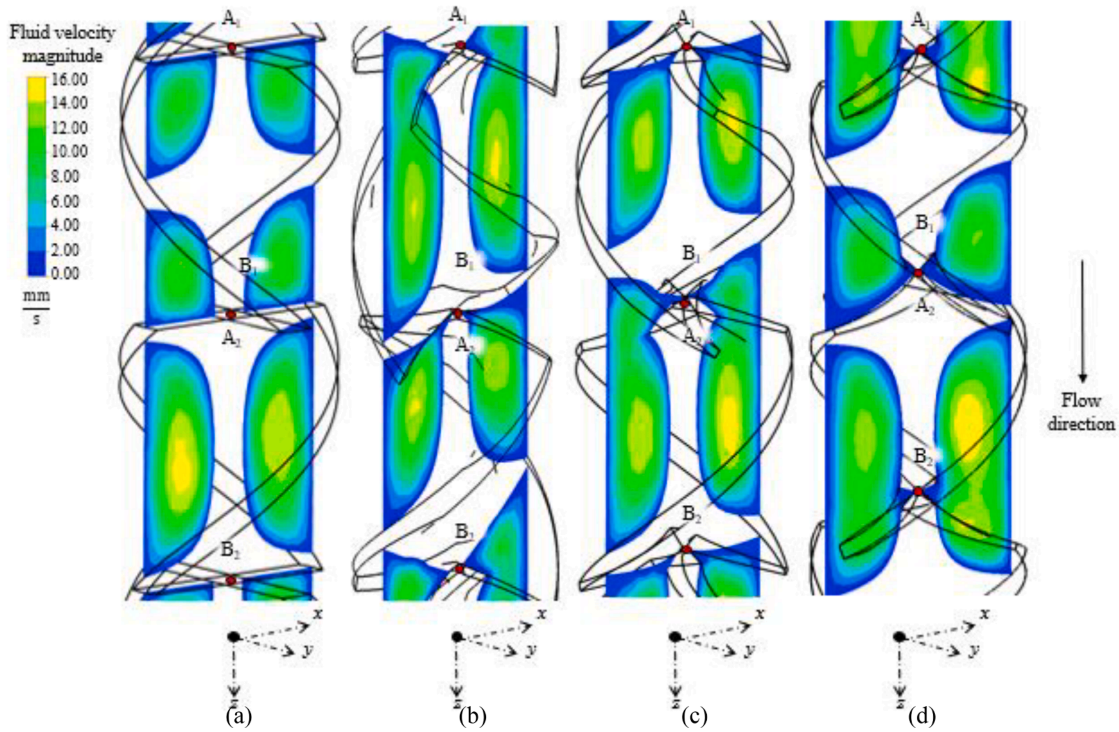

Fig. 16. Velocity contour plots on the central *x-z* plane of the (a) standard SSM, (b) 15-15-22.5, (c) 22.5-22.5-22.5, and (d) 30-30-22.5 Impeller SSMs. The three Impeller SSM configurations have a larger and more continuous high fluid velocity regions than those of the standard SSM.

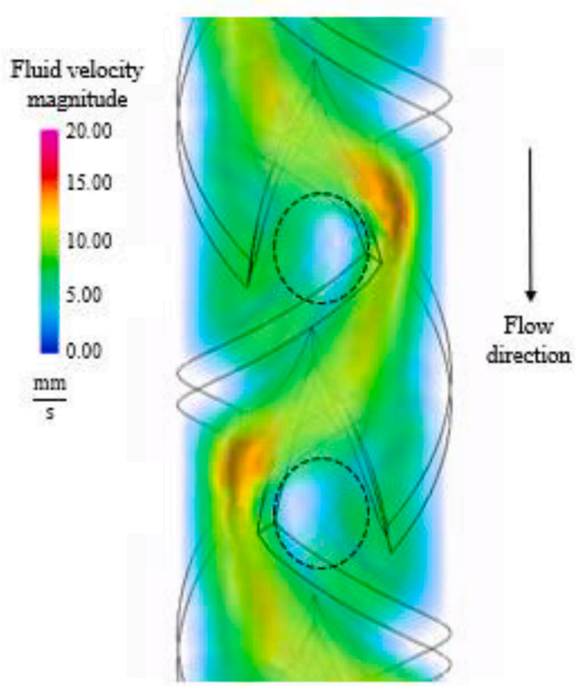

Fig. 17. Stagnation point created by the release winglet and taper inlet of the 67.5-67.5-22.5 Impeller SSM.

designs. Another area of future work is investigating the flow fields within the AM static mixer. The flow within the SSM is three dimensional and 3D flow field using stereoscopic techniques would provide more data on the performance of the mixers, this was not attempted with the ISSM due to the opaqueness and size of the design.

Table 6
K_L for various Static mixer types.

Mixer	K_L
Standard SSM	6.9
ISSM	5.9
SMX	37.5
SMXL	7.8
SMR	46.9

CRediT authorship contribution statement

Matthew Hildner: Conceptualization, Methodology, Writing – original draft, Data curation, Investigation, Formal analysis, Methodology. **James Lorenz:** Investigation. **Bizhong Zhu:** Supervision, Resources. **Albert Shih:** Supervision, Project administration, Writing – review & editing, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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